AareML — Project Effort Log

CAS in Advanced Machine Learning · University of Bern

Total budget: 120 hours (4 ECTS) · Deadline: 15 June 2026

138.0	−18.0	115%
Hours Logged	Hours Remaining	Complete

Effort Log

Session	Date	Activity	Category	Hours	Cumul.
50	01 Jun 2026	nb18 results analysis + report/readme/effort log update	Writing	1.5	4.0
49	31 May 2026	Innovation Sandbox interview prep + pitch deck (6 slides)	Writing	2.0	6.0
48	28 May 2026	NeuralHydrology nb17 + nb04c/nb18 UBELIX debugging	Engineering	3.0	11.0
47	27 May 2026	nb18 cascaded physics-informed model (professor suggestion)	Modelling	2.0	15.0
46	27 May 2026	3-model peer review + code review — 8 critical fixes applied	Engineering	3.5	19.0
45	19 May 2026	New presentation rebuild (13 slides) + repo cleanup	Writing	2.0	23.0
44	19 May 2026	Personal presentation script (15 min narrative)	Writing	2.5	26.0
43	18 May 2026	Q&A; training Day 9 continued + literature deep dive	Writing	2.0	29.0
42	17 May 2026	PFAS research — Swiss water data landscape	Planning & Research	1.5	30.5
41	17 May 2026	Q&A; training Day 8-9 (mock professor questions)	Writing	2.0	33.5
40	16 May 2026	Short lookback ablation (4-7 days) + report v1.25	Modelling	2.0	36.5
39	16 May 2026	Innovation Sandbox v3 (Eawag partnership strategy)	Planning & Writing	2.0	39.5
38	16 May 2026	Literature review (Zhi 2023, Padrón 2025, Baste 2025)	Planning & Research	2.5	42.5
37	15 May 2026	CAMELS-CH base attributes (Höge 2023) integration	Modelling	3.0	46.5

Session	Date	Activity	Category	Hours	Cumul.
36	14 May 2026	Presentation script + slides update	Writing	2.0	51.5
35	13 May 2026	Ridge zero-shot + nb16 cross-validation design	Modelling	3.0	54.5
34	12 May 2026	Presentation training Day 1-3 (LSTM, training, transfer)	Writing	2.5	57.5
33	11 May 2026	UBELIX reruns (Q bug fix, nb04/04b/15)	Infrastructure	3.0	60.5
32	8 May 2026	Scientific restraint rewrites v1.23	Writing	2.0	64.5
31	7 May 2026	nb04c EC precipitation proxy	Modelling	2.0	67.0
30	6 May 2026	EA-LSTM temperature (nb04b) + 86-gauge expansion	Modelling	3.0	69.5
29	5 May 2026	CAMELS-CH static attributes integration	Modelling	2.5	71.5
28	1 May 2026	Thiago Nascimento feedback — EA-LSTM updates	Modelling	3.0	74.5
27	23 Apr 2026	nb15 Scientific rigor (Granger, ACF, threshold)	Modelling	2.5	76.5
26	22 Apr 2026	nb14 AR baseline + professor feedback Q&A;	Modelling	2.0	79.0
25	21 Apr 2026	nb12 Error analysis + nb13 Seasonal analysis	Modelling	2.5	81.0
24	20 Apr 2026	nb11 Ablation study	Modelling	2.0	83.5
23	19 Apr 2026	Innovation Sandbox application draft	Writing	3.0	86.5
22	18 Apr 2026	Professor feedback integration v1.22	Writing	2.0	89.0
21	17 Apr 2026	Fact-check & corrections v1.21	Writing	2.5	92.0
20	16 Apr 2026	Report shortening v1.19 + TOC v1.20	Writing	2.5	94.0
19	15 Apr 2026	UBELIX reruns (seed bug, EA-LSTM scaler fix)	Modelling	4.0	96.0
18	14 Apr 2026	Presentation design (PPTX, fish boxes)	Writing	3.0	99.0
17	13 Apr 2026	Results integration & report v1.11-v1.15	Writing	3.0	101.5
16	12 Apr 2026	UBELIX HPC setup & job submission	Infrastructure	3.0	104.5
15	11 Apr 2026	Full code audit & bug fixes (3-model review)	Engineering	5.0	106.5
14	10 Apr 2026	Report writing v1.0-v1.10	Writing	4.0	109.5
13	9 Apr 2026	Swiss lakes LSTM (nb10)	Modelling	3.0	112.0
12	9 Apr 2026	Canton Zurich analysis (nb09)	Modelling	3.0	114.0

Session	Date	Activity	Category	Hours	Cumul.
11	8 Apr 2026	USGS cross-continental transfer (nb08)	Modelling	3.0	116.0
10	8 Apr 2026	Cross-ecosystem transfer (lakes nb06, nb07)	Modelling	3.0	118.0
9	7 Apr 2026	GitHub upload & environment setup	Infrastructure	1.5	119.5
8	7 Apr 2026	Report writing	Writing	3.0	121.5
7	6 Apr 2026	Multi-site & SHAP notebooks	Modelling	3.0	124.0
6	6 Apr 2026	Code improvements & refactoring	Engineering	4.0	126.0
5	6 Apr 2026	LSTM implementation	Modelling	4.0	129.5
4	6 Apr 2026	Baseline models	Modelling	4.0	131.5
3	5 Apr 2026	Data acquisition & EDA	Data & EDA	5.0	134.5
2	5 Apr 2026	Project definition & proposal	Planning & Writing	2.0	136.5
1	17 Mar 2026	Project scoping & dataset research	Planning & Research	4.0	138.0
		TOTAL		138.0	138.0

Planned Work

Task	Est.
Run notebook 03 full Optuna tuning	3 h
Run notebook 04 multi-site evaluation	2 h
Run notebook 05 SHAP attribution	2 h
Update report with full LSTM + multi-site + SHAP results	4 h
Discussion sections 6.3–6.4	2 h
Final polish, figures, proofreading	3 h
GitHub final push + submission	1 h
Buffer / unexpected issues	10 h
Total planned remaining	27 h + buffer

Notes

- Hours are rounded to the nearest 0.5 h.
- Sessions may span multiple calendar days; the date shown is the primary work date.
- Budget includes all project work: planning, coding, writing, and infrastructure.
- ECTS conversion: 4 ECTS = 120 hours (30 h / ECTS).